

Program : Diploma in Wood and Paper Technology	
Course Code : 3123	Course Title: Wood Processing
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide exposure on various processing methods of Wood and wood based raw materials.
- To impart basic knowledge on felling, conversion, seasoning and preservation.
- To provide basic idea on biological deterioration and its causes and remedies.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Knowledge of basic wood science		Botany and Anatomical Structures	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the forest, forestry and forest operations	15	Understanding
CO2	Summarize the felling and transportation of logs	14	Understanding
CO3	Explain the causes and remedies of attack by biological agencies and need of wood preservation	15	Understanding
CO4	Interpret the seasoning of wood	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2	3						
CO3	3						
CO4			3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the forest, forestry and forest operations		
M1.01	Recall types of forest and their distribution in India	3	Remembering
M1.02	Explain the role of forest - protective role, productive role and bio - aesthetic role	4	Understanding
M1.03	Explain Forest operations	5	Understanding
M1.04	Discuss Logging practices in India	3	Understanding
Contents: Forest, forestry, forest products, trees, type of forest and their distribution in India - tropical wet evergreen forest, semi ever green forest, moist deciduous forest, dry evergreen forest, distribution of bamboos and reed forests in India. Role of forest - protective role, productive role and bio-aesthetic role. Farm forestry. Forest operations - working plan - structure of forest management, Silvi cultural systems - clear felling systems - reforestation plantations - intensive plantation and trench mound plantations. Logging - logging practices in India, logging waste how to avoid it. Implements used in felling and conversion			
CO2	Summarize the felling and transportation of logs		
M2.01	Explain the principles of felling and methods of felling.	4	Understanding
M2.02	List the classification of trees	3	Understanding
M2.03	Explain identification of timbers	4	Applying
M2.04	Identify defects in timber	3	Understanding
	Series Test - I	1	

Contents:**Felling, Classification and Identification of timber**

Felling - principles of felling, rules of felling, methods of felling dry storage wet storage. Water storage, Calculation of log volume by Hoppus Method, Off- road and on -road transportation- Classification of trees - exogenous and exogenous trees. Cross-section of exogenous tree, comparative study of hardwood and soft wood. Utilization of small log, important timbers - characteristics and uses of timbers.

Natural defects - causes and features of natural defects with figure - knots check, split, shakes, cross grain, discoloration, compression wood, tension wood, pith pocket identification of timber - dichotomous keys - perforated card key - key to important timbers and identification - identification, properties and description.

CO3	Explain the causes and remedies of attack by biological agencies and need of wood preservation		
M3.01	Discuss biological deterioration of wood	4	Understanding
M3.02	Explain Wood Preservation	4	Applying
M3.03	Describe the mechanical properties of wood	3	Understanding
M3.04	Explain the methods and application of Preservatives	4	Applying

Contents:**Biological Deterioration, Mechanical properties and Preservation-**

Biological deterioration of wood - Brown rot white rot, soft rot, mould rot- sap stain-physiological requirement of wood rotting fungus wood inhibiting fungi. Wood boring insects white ant marine borers powder beetles, carpenter ants and their deteriorative action on wood

Mechanical properties of wood -Review of stress strain diagram - behavior of mild steel & brittle material under tension and identify the significant points-limit of proportionality - elastic limit - yield point) - ultimate stress - working stress - factor of safety - Hooks law and Young's modulus, tensile strength, impact strength, hardness, compressive strength, bending strength, shear strength screw and nail holding power.

Shear stress and shear strain - modulus of rigidity - volumetric strain- Young's Modulus (E)- Bulk Modulus (K) and Modulus of Rigidity (C) relations-simple problems

Types of beams - cantilever beam- simply supported beam- over hanging beam- built in beam or fixed beam and continuous beam - types of loading - concentrated or point load- uniformly distributed load and uniformly varying load -

Wood preservation - define preservation - need of preservation - objects of preservation. Requirements good preservative. Types of preservatives - Oil type, water soluble type, organic solvent type. Coal tar creosote, ASCU, CCA, CCB, Zinc chloride, Zinc Meta arsenite, boric acid, borax powder, and sodium pentachlorophenate.

Methods of Preservation - non-pressure process and pressure process; Brushing, spraying dipping, steeping, osmos process, hot and cold bath, diffusion process, boucherie and modified boucherie process

CO4	Interpret the Seasoning of wood		
M4.01	Recall seasoning of wood	4	Understanding
M4.02	Explain Seasoning Methods	5	Understanding
M4.03	Interpret Seasoning Defects, explain buckling load on given wooden column	5	Applying
	Series Test - II	1	

Contents:

Seasoning of timber- Seasonings of wood -.Advantages of seasoned wood. Shrinkage and swelling of wood. Fiber saturation point. Equilibrium moisture content. Free moisture and bound moisture classification of timber with respect to seasoning purpose - refractory, non-refractory and moderately refractory timbers.

Seasoning methods - natural and artificial - natural (Air) seasoning, water seasoning, chemical seasoning, solar seasoning - electrical seasoning Kiln seasoning - advantageous and disadvantageous of Kiln seasoning. Types of kiln - Progressive and compartment type. Stacking methods in kiln . Kiln schedules, wet bulb and dry bulb. State the Need of vacuum drying - calculate moisture content by oven - dry method.

List the Seasoning defects - draw the defects - twist, spring, cup, bow, cross shakes, honey combing, surface checks, splitting causes and remedies of the above defects.

Introduction - column - strut - buckling load - equivalent length - slenderness ratio - types of columns - short column - medium size column - long column - Euler's equations and its assumption for crippling load for different end conditions (no proof)

Text / Reference:

T/R	Book Title/Author
T1	Franz F. P. Kollmann ,Principle of wood science & technology Vol. I - Springer
T2	Barry A. Richardson, Wood preservation - construction press London
T3	Dr. S.S Negi, Forests and Forestry - National Book trust , India
R1	Forest Utilization - volume I & II - F R I, Dehradun
R2	Lecture notes on plywood technology volume I - IPIRI, Bangalore